DataNova: Automated EDA Report

Format: Executive | Domain Detected: Healthcare

Executive / Dataset Overview

The dataset captures a snapshot of 309 patients and includes 16 variables that describe basic demographics (gender, age) and a range of health-related indicators such as smoking status, symptom presence (e.g., wheezing, chest pain), and lifestyle factors (alcohol consumption). The primary business question is whether these attributes can help identify individuals at risk for lung cancer, as indicated by the "LUNG_CANCER" column. The modest size of the data set makes it well-suited for rapid prototyping of predictive models that could support early-detection initiatives or targeted screening programs.

Data Quality & Profiling Insights

Overall data health is strong: there are no missing values and data types are consistent across columns. The schema is clean, though a few column names contain trailing spaces (e.g., "FATIGUE ", "ALLERGY ") which could cause downstream processing issues and should be standardized.

The only notable anomaly is in the age distribution. Two records (rows 22 and 238) fall below the statistical lower bound for age (39 years) with values of 21 and 38 years. These outliers represent less than 1 % of the sample and are unlikely to distort aggregate insights, but they should be reviewed to confirm that they are genuine cases rather than data-entry errors.

All other variables are binary or categorical integer flags (0/1) and appear to be recorded uniformly, supporting straightforward aggregation and modeling.

Strategic / Operational Takeaways

1. **Readiness for Predictive Analytics** - The clean, complete data set can be leveraged immediately to develop a risk-scoring model for lung cancer. Early-stage models can be validated against the existing "LUNG_CANCER" labels to assess predictive power and guide resource allocation for screening programs.

2. **Data Governance Actions** - Standardize column names by removing trailing spaces and adopting a consistent naming convention (e.g., snake_case). Document the rationale for each variable to ensure alignment with clinical terminology and facilitate future data integration.

3. **Outlier Review Process** - Implement a quick validation step for age outliers. If confirmed as valid cases, they may highlight younger patients with atypical risk profiles, which could be valuable for refining risk thresholds. If they are errors, correct or exclude them to preserve model integrity.

4. **Feature Enrichment Opportunities** - Consider augmenting the data with additional clinical variables (e.g., family history, imaging results) or socioeconomic factors that are known to influence lung cancer risk. This will improve model robustness and broaden the strategic insights that can be derived.

5. **Operational Impact** - A reliable risk model derived from this data can enable targeted outreach, prioritize high-risk patients for low-dose CT scans, and ultimately reduce diagnostic delays. The concise data set also allows for rapid iteration, meaning the organization can move from prototype to pilot within weeks rather than

months.

In summary, the dataset is of high quality and ready for immediate analytical use. Minor housekeeping-cleaning column names and confirming age outliers-will further strengthen the foundation for building actionable lung-cancer risk tools that support early detection and more efficient allocation of clinical resources.

Visual Analysis